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Using Lasso for Predictor Selection and to Assuage Overfitting: A Method Long Overlooked in Behavioral Sciences

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Ordinary least squares and stepwise selection are widespread in behavioral science research; however, these methods are well known to encounter overfitting problems such that R^2 and regression coefficients may be inflated while standard errors and p values may be deflated, ultimately reducing both the parsimony of the model and the generalizability of conclusions. More optimal methods for selecting predictors and estimating regression coefficients such as regularization methods (e.g., Lasso) have existed for decades, are widely implemented in other disciplines, and are available in mainstream software, yet, these methods are essentially invisible in the behavioral science literature while the use of sub optimal methods continues to proliferate. This paper discusses potential issues with standard statistical models, provides an introduction to regularization with specific details on both Lasso and its related predecessor ridge regression, provides an example analysis and code for running a Lasso analysis in R and SAS, and discusses limitations and related methods.

KEYWORDS lasso; overfitting; regularization; regression

In behavioral science research, substantive conclusions based upon statistical analyses are ubiquitous across disciplines. A large majority of quantitative studies focus on some form of regression such that a set of predictor variables (i.e., independent variables) are hypothesized to explain (although not necessarily causally) an outcome or set of outcomes (i.e., dependent variables). Many diverse types of statistical methods are extensions of the basic regression problem of predicting an outcome from another variable or set of variables including structural equation modeling, multilevel models, or generalized linear models.

Although the general setup of regression problems is intuitive and straightforward, in practice, the process of selecting relevant predictor variables to include in the model is a challenging endeavor that is highly influential for drawing conclusions from the model. In fact, the omnipresent nature of regression models has led prominent statisticians such as Bradley Efron to identify predictor variable selection as the single most important area of statistical research

(Hesterberg, Choi, Meier, & Fraley, 2008). In concordance with this stance, a great deal of the recent statistical literature has focused on developing and refining modern statistical methods to better select predictor variables in regression models and to better estimate effects (again, not necessarily causal) that predictors have on an outcome. Arguably, the most widely researched and implemented modern method is the least absolute shrinkage and selection operator (Lasso) first developed by Tibshirani (1996), which fits within the broader least angle regression framework (LARS; Efron, Hastie, Johnstone, & Tibshirani, 2004) that can estimate Lasso with the computational complexity of ordinary least squares (OLS). These methods are extremely widespread in both statistical applications and the statistical literature, as evidenced by the seminal paper by Tibshirani (1996) having over 11,000 citations and Efron et al. (2004) having over 4,500 citations, according to Google Scholar at the time of this writing.

Despite the overwhelming support for modern methods like Lasso in the statistical literature and in statistical applications in other disciplines (biomedical fields in particular), behavioral sciences have largely relied on more traditional methods such as p values or automatic selection methods such as forward, backward, or stepwise selection to

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determine which predictors to retain in the model. As informal supporting evidence, a Google Scholar search for “stepwise regression” limited to psychology or education journals since 2010 results in over 1,000 results, even though shortcomings of these methods have been disseminated for over a decade (e.g., Harrell, 2001; Hurvich & Tsai, 1990; Johnson & Omland, 2004; Knapp & Sawilowsky, 2001; Stephens, Buskirk, Hayward, & Martinez del Rio, 2005; Steyerberg, Eijkemans, & Habbema, 1999; Thompson, 1995, 2001; Whittingham, Stephens, Bradbury, & Freckleton, 2006; Wintle, McCarthy, Volinsky, & Kavanagh, 2003). Moreover, articles employing these methods were not strictly limited to lower-tier, low-impact journals—flagship journals in these areas such as *Child Development* and *Developmental Psychology* each published at least 10 articles using stepwise selection since 2010 (granted, this does not represent an overwhelming proportion of the 500+ studies published in each of these publications in that timeframe). Furthermore, no articles in education journals and only a handful of studies (e.g., Hartmann, Van Der Kooij, & Zeeck, 2009; Hartmann, Zeeck, & Barrett, 2010; Scheidt et al., 2012; Schmid, Taylor, Foldi, Berres, & Monsch, 2013) in psychology journals outside of neuroscience reported using Lasso in a literature search from Google Scholar and PsycInfo. Although stepwise selection is not completely without merit, there is clearly a sizeable disconnect between the statistical methods implemented in behavioral science applications and advances in the statistical literature, even though modern methods such as Lasso are widely available in software packages familiar to behavioral science researchers such as R, SAS, and SPSS.

Following suggestions from Sharpe (2013), which discussed the need for more methodological papers that bridge advances in statistical methodology to applied researchers who utilize statistical methods, the goal of this paper is to introduce and illustrate Lasso to behavioral science researchers. A paper of this nature seemed particularly worthwhile for Lasso because nearly all research on Lasso appears in journals that emphasize more theoretical treatments of statistical methodology (e.g., *Annals of Statistics*, *the Journal of the American Statistical Association*, *the Journal of the Royal Statistical Society, Series B*), which behavioral science researchers may not read frequently or may not have an interest or background in theoretical statistical research.

This paper first overviews traditional methods such as OLS and stepwise selection and the issues associated with them in common behavioral science contexts. Second, the general conceptual idea of regularization methods (the broader class of methods to which Lasso belongs) is introduced. Third, Lasso and its conceptual predecessor, ridge regression, will be discussed in detail. Then, an applied example is shown using an educational data set from Stock and Watson (2003) that predicts standardized test scores for school districts in California. This example emphasizes how to conduct a Lasso analysis in R, which currently has the most advanced procedures for Lasso. SAS code is also pro-

vided in the appendix. Lastly, limitations and extensions of Lasso and its implementation in software are discussed.

OLS REGRESSION AND POTENTIAL ISSUES

OLS regression is a frequently applied method to predict a metric outcome from many predictors. In standard OLS regression, coefficient estimates are obtained by minimizing a loss function such that the vertical squared distances between the observed data values and the predicted values are the smallest. Mathematically, this loss function can be formulated as $L^{\text{OLS}}(\beta) = \|\mathbf{Y} - \mathbf{X}\beta\|^2$ where L is the loss function, $\mathbf{X}$ is an $n \times p$ design matrix of predictors, $\mathbf{Y}$ is an $n \times 1$ vector of responses, and β is a $p \times 1$ vector of regression coefficients where n is the number of observations¹ and p is the number of predictors (including the intercept). The multiplication of $\mathbf{X}\beta$ produces an $n \times 1$ vector of predicted values that is often denoted as $\hat{\mathbf{Y}}$. With OLS, the estimates for the vector of regression coefficients β have an algebraic, closed-form solution such that

$$\hat{\beta}^{\text{OLS}} = (\mathbf{X}^T \mathbf{X})^{-1} (\mathbf{X}^T \mathbf{Y}) \quad (1)$$

When fitting any type of regression model, random noise can become entangled with signal, especially with small or moderate sample sizes. This can lead to estimates that overstate the impact of particular predictor variables and attribute more predictive power to the model than is present in the population. When this occurs, the model is said to be overfit with the consequence that regression coefficients have inflated magnitude, standard errors are underestimated, p values are consequently too small, R^2 values are consequently too large compared to their population values, and the model is not parsimonious because extraneous predictors may be seen as important (Babak, 2004; Cohen, Cohen, West, & Aiken, 2003; Derksen & Keselman, 1992). Each of these quantities are vital to model interpretation and their properties are increasingly misleading as the ratio of the number of observations (n) to the number of predictors (p) decreases (Babak, 2004; Derksen & Keselman, 1992; Hesterberg et al., 2008). It is important to note that overfitting is not the result of biased estimates. Estimates from an overfit model do perform well for the sample data from which they were fit; however, they perform far less admirably when applied to a different sample from the same population, which hurts the generalizability of conclusions drawn from the model. Furthermore, if a fairly large number of candidate predictors

¹ $\|\mathbf{Y} - \mathbf{X}\beta\|$ is the ℓ_2 norm of the difference between the observed values $\mathbf{Y}$ and the predicted values $\mathbf{X}\beta$. In linear algebra, the ℓ_2 norm (a.k.a. the Euclidean norm) is square root of the summation of the argument. That is,

$\|\mathbf{Y} - \mathbf{X}\beta\| = \sqrt{\sum_i [\mathbf{Y}_i - (\beta_0 + \sum_{p=1}^P \beta_p \mathbf{X}_{ip})]^2}$. Squaring the ℓ_2 norm is simply a notationally convenient way to write the sum of squared residuals in matrix form.

exist and model selection is of interest, the issue of overfitting can be further exacerbated because inference in regression models makes the assumption that the predictors are fixed entities, but methods like stepwise regression repeatedly add and subtract predictors to the model without accounting for this adaptive nature of the model (Lockhart, Taylor, Tibshirani, & Tibshirani, 2014; Thompson, 2001).

When using model selection procedures (e.g., forward, backward, stepwise regression), problems associated with overfitting become magnified and t statistics and F statistics used for inference do not follow their appropriate null distributions nor do they have appropriate degrees of freedom (Pope & Webster, 1972; Thompson, 1989, 1995). That is, standard inferential tests and their associated p values are not equipped for repeatedly swapping predictors in and out of the model, all while significance testing along the way. Thus, the operating Type I error rates for hypothesis tests on the regression coefficients are too large, leading to more variables being included in the model than may be true in the population and intensifying the effect of overfitting (Wilkinson, 1979).

WHY ARE MODELS OVERFIT?

$\hat{\beta}^{\text{OLS}}$ is the best linear unbiased estimator (BLUE), so it is important to note that overfitting is not a problem of biased estimates. However, the bias-variance tradeoff must be considered. Briefly, the bias-variance tradeoff describes the dilemma whereby unbiased estimates are appropriate for the current, seen data, yet unbiased estimates are more susceptible to minor changes or fluctuations in unseen data from different samples or that will be collected in the future. In other words, the estimates that are unbiased and minimize the prediction error (as OLS does) for a single sample have more variance between samples. Conversely, biased estimates are not ideal for a single seen dataset; however, biased estimates may have less variance from sample to sample provided that the bias is not overwhelmingly large. Considering the bias-variance tradeoff, OLS estimates are unbiased² but more variable from sample to sample, which explains why R^2 values found in one dataset tend to not generalize well to other data sets from the same population. That is, an R^2 value based on OLS estimates will be higher, almost universally, in the sample used to fit the model compared to an R^2 obtained from using the same regression coefficients in another sample from the same population.

As an extreme example, consider a regression model that has an identical number of predictors as observations (i.e., a saturated model). The fit of this saturated model will be perfect: the predicted values will be exactly equal to the observed

values and R^2 will be exactly 1. However, if more data were collected and the outcome is predicted from the estimated coefficients fit to the original data, the model would likely not fit nearly as well to the second sample as it did to the first.

As a more general hypothetical example, consider two samples drawn from the same population, Sample A and Sample B. An OLS regression model is first fit to Sample A. Because OLS is the BLUE, the estimates will be unbiased (provided assumptions are met) and the prediction errors will be minimized for Sample A. However, to the extent that Sample A differs from Sample B, the regression coefficients estimated from Sample A will perform less well when applied to Sample B. Additionally, again to the extent that Sample B differs from Sample A, the regression coefficients for an OLS model fit to Sample B will be different from those estimated from Sample A (i.e., the estimates will have between-sample variance). Keeping in mind that the goal of a regression model is to make inferences about population effects, the higher variance of the coefficient estimates can make this inferential process more difficult, especially in real-life contexts where there is only a single sample from which estimates are obtained and the variance from sample to sample cannot easily be ascertained. As discussed in the next section, incorporating some controlled amount of bias into the estimates may help the generalizability of estimates and conclusions drawn from them. That is, even though the coefficient estimates will no longer be the BLUE for the sample of data used to fit the model, they will be more appropriate across a range of samples and, hopefully, to the population.

Although potential issues with overfitting are not new to many behavioral science researchers and these ideas have been circulated throughout many literatures (see, e.g., Andersen & Bro, 2010; Babyuk, 2004; Cohen et al., 2003; Harrell, 2001; Harrell, Lee, Matchar, & Reichert, 1985; Hawkins, 2004), methods that are susceptible to overfitting continue to be extensively applied in behavioral science analyses. The following sections discuss alternative methods that address the same research questions as OLS regression but that take overfitting concerns into account.

IDEA BEHIND REGULARIZATION

Regularization is one method that is used to account for issues of overfitting with OLS (as well as models estimated with maximum likelihood like logistic regression, Poisson regression, etc.). Rather than minimizing the loss function directly as in OLS, regularization methods use a penalized loss function such that

$$L^{\text{Reg}}(\beta) = L^{\text{OLS}}(\beta) + \lambda P(\beta) \quad (2)$$

where L^{Reg} is the penalized loss function, L^{OLS} is the standard OLS loss function, λ is a regularization parameter (aka tuning parameter), and P is a penalty function that varies

²OLS estimates are unbiased provided that model assumptions are correct such as homoscedasticity and normality of the residuals and proper specification of the model in terms of including the appropriate predictors with the appropriate functional form.

depending on the type of regularization being applied (e.g., ridge, Lasso, elastic net). Because the loss function is minimized, adding a penalty term to L^{OLS} shrinks the magnitude of the coefficient estimates, reducing coefficient estimates and, consequently, overfitting. The extent of the penalization is determined by λ . More specifically, as $\lambda \rightarrow 0$ the resulting coefficient estimates are equivalent to OLS estimates because the penalty term is reduced to 0 and penalized loss function reduces to the OLS loss function. As $\lambda \rightarrow \infty$, the penalty becomes extremely large and estimates approach zero or may equal zero, depending on the penalty function as will be discussed shortly. Put another way, regularization still minimizes $L^{\text{OLS}}(\beta)$ but under some constraint, which prevents the coefficient estimates from having inappropriately large magnitudes. Determining how much or how little to penalize the loss function is vitally important with regularization methods because it is important to penalize the loss function enough to reduce overfitting but not too much so that estimates are rendered meaningless from excessive bias. Selecting a value for λ will be discussed in detail in a separate section.³

Reverting back to the bias-variance tradeoff, regularization methods purposefully introduce bias to the coefficient estimates (by shrinking them towards 0). Although this seems undesirable intuitively, recall that overfitting can result from estimates capitalizing on potential idiosyncrasies of an individual sample of data. From the bias-variance tradeoff, through the controlled bias introduced through regularization, the variance of the estimates is reduced so that estimates are more stable, less susceptible to small changes in the data, and thus more generalizable to data other than those used to fit the model.

Although the discussion up to this point has focused on OLS, the ideas can be naturally extended to models estimated with maximum likelihood (such as generalized linear models) with a few small adjustments. Namely, the OLS loss function is replaced with the likelihood function or, more plausibly, the log-likelihood function. Additionally, as implied by the name, because maximum likelihood is a maximization problem rather than minimization problem as with OLS, the penalty term would be subtracted rather than added. To maintain a targeted narrative, the remaining sections will focus on OLS, but readers should be cognizant that the methods discussed are not limited only to analytic scenarios that are appropriate for OLS and are easily extended to models estimated with maximum likelihood.

³In physics and numerical mathematics where regularization originated, regression model fitting can be considered an inverse problem because the data are collected first and parameters are estimated afterwards based on the collected data. Furthermore, model fitting is often an ill-posed problem because the parameters are sensitive to small fluctuations in the data (i.e., overfitting) or problems may have multiple or zero solutions (Hadamard, 1902). Regularization introduces additional information into the problem (through a penalty term) that addresses the ill-posed nature of model fitting by reducing the sensitivity to small data fluctuations (Ambartsumian, 1929).

Regularization and Bayes

Although regularization is not necessarily conducted within a Bayesian framework, a conceptual link between regularization and Bayesian methods can be made (Gelman & Shalizi, 2013). As a quick overview of Bayesian methods, parameters to be estimated in a Bayesian framework are viewed as random variables that are conditional on the data. This is in opposition to the frequentist view, which considers the parameters to be fixed but unknown quantities and the data to be random. Since parameters are considered random variables in Bayesian analyses, parameters have distributions rather than being estimated with a single value as in a frequentist framework. Each parameter has a posterior distribution, which is often summarized with some measure of central tendency (i.e., mean, median, or mode) and serves as the “point estimate” for that particular parameter. The posterior distribution in a Bayesian context is a combination of the likelihood as calculated from the data (the same likelihood as in a maximum likelihood problem in a frequentist framework) and a prior distribution that is specified by the researcher a priori for each parameter. The prior distribution can capture knowledge from the researcher and/or previous studies, although the prior can, and often is, set to be rather uninformative, giving more credence to the data in the calculation of the posterior distribution. As prior distributions become more informative, they tend to “pull” estimates towards them because the posterior is a combination of the likelihood and the prior. In practice, priors for regression coefficients are often centered around zero, meaning that estimates are shrunk towards zero and increasingly so as the prior becomes increasingly informative.

Regularization has the same type of effect, although it operates through a different mechanism. Regularization operates under the notion that regression coefficients are more plausibly equal to zero because standard frequentist methods tend to incorporate some noise or sampling variability into the regression coefficients (i.e., if a regression coefficient is null in the population, its estimate from a sample will almost certainly be non-null). Similar to prior distributions in Bayesian methods, the penalty function shrinks the coefficient estimates towards zero so that they are more consistent with this expectation.⁴ Also similar to Bayesian methods, the degree of the shrinkage (which in Bayes is determined by the informativeness of the prior) is determined by λ where large values heavily shrink estimates towards 0 (similar to a highly informative Bayesian prior) and small values shrink estimates more moderately (similar to a weakly informative Bayesian prior). Procedures for selecting a value for λ will be discussed in the next section as well as in the context of the provided applied example later on.

⁴Regularization is not typically applied to the intercept because the intercept is not typically considered to be affected by overfitting in a similar manner as regression coefficients.

OVERVIEW OF REGULARIZATION METHODS

Ridge Regression

Although the focus of this paper is on Lasso, ridge regression (Hoerl & Kennard, 1970; Tikhonov, 1943) is a natural predecessor of Lasso that provides a foundation for discussing Lasso and also maintains some advantages in its own right that are worth mentioning. As noted previously, regularization methods apply a penalty function $P(\boldsymbol{\beta})$ that shrinks coefficient estimates towards 0. In ridge regression, the penalty applies a constraint on the sum of the squared regression coefficients such that $P^{\text{Ridge}}(\boldsymbol{\beta}) = \lambda \sum_{j=1}^p \beta_j^2$. Consistent with the matrix notation provided previously, this corresponds to a penalized loss function equal to

$$L^{\text{Ridge}}(\boldsymbol{\beta}) = \|\mathbf{Y} - \mathbf{X}\boldsymbol{\beta}\|^2 + \lambda \boldsymbol{\beta}^T \boldsymbol{\beta} \quad (3)$$

where L is the loss function, $\mathbf{X}$ is an $n \times p$ design matrix for predictors, $\mathbf{Y}$ is an $n \times 1$ vector of responses, $\boldsymbol{\beta}$ is a $p \times 1$ vector of regression coefficients, and $\lambda \geq 0$ is the regularization parameter that controls the degree of shrinkage. Using the sum of squared regression coefficients (aka an ℓ_2 penalty) such that $P^{\text{Ridge}}(\boldsymbol{\beta}) = \lambda \sum_{j=1}^p \beta_j^2$ results in shrinkage that is proportional to the size of the coefficient: the larger the coefficient estimate, the greater the shrinkage will be (Hesterberg et al., 2008).

Estimates of $\boldsymbol{\beta}$ are obtained by minimizing $L^{\text{Ridge}}(\boldsymbol{\beta})$; which has a closed-form solution such that

$$\hat{\boldsymbol{\beta}}^{\text{Ridge}} = (\mathbf{X}^T \mathbf{X} + \lambda \mathbf{I})^{-1} \mathbf{X}^T \mathbf{Y} \quad (4)$$

where $\mathbf{I}$ is an identity matrix of dimension p .

Comparing Equation (4) to Equation (1), it can be seen fairly readily that as $\lambda \rightarrow 0$ in Equation (4), the ridge estimates approach OLS estimates shown in Equation (1). However, as $\lambda \rightarrow \infty$ estimates will not be shrunk to exactly zero meaning that ridge regression does not perform de facto predictor selection—all predictors are retained in the model (i.e., ridge regression does not “zero out” certain predictors; Lasso does have this property as will be discussed in the next section). As a benefit of model selection not being performed in ridge regression, standard errors can be calculated by taking the square root of the diagonal terms of $\text{Var}(\hat{\boldsymbol{\beta}}^{\text{Ridge}}) = \sigma^2 (\mathbf{X}^T \mathbf{X} + \lambda \mathbf{I})^{-1} \mathbf{X}^T \mathbf{X} (\mathbf{X}^T \mathbf{X} + \lambda \mathbf{I})^{-1}$ and standard inference can be performed so long as the appropriate OLS assumptions are met because the model is not adaptive and the predictors are truly fixed entities.

As an additional benefit, ridge regression is better equipped to handle collinear predictors because of the $\lambda \mathbf{I}$ ridge term that appears in $\hat{\boldsymbol{\beta}}^{\text{Ridge}}$ but not in $\hat{\boldsymbol{\beta}}^{\text{OLS}}$. With OLS and highly collinear predictors (i.e., when $\mathbf{X}$ is not of full rank), $(\mathbf{X}^T \mathbf{X})^{-1}$ may not be invertible meaning $\hat{\boldsymbol{\beta}}^{\text{OLS}}$ will not be calculable. In ridge regression, the additional $\lambda \mathbf{I}$ ridge term in $\hat{\boldsymbol{\beta}}^{\text{Ridge}}$ ensures that $(\mathbf{X}^T \mathbf{X} + \lambda \mathbf{I})^{-1}$ will also be invertible,

avoiding situations that are inestimable with OLS (Vinod, 1978).

Lasso

Lasso was first introduced by Tibshirani (1996) and is conceptually similar to ridge regression in that it utilizes a penalty that operates as a function of the regression coefficients. Rather than minimizing the loss function by constraining the sum of squares for the coefficients as in ridge regression (an ℓ_2 penalty), Lasso sets a constraint on the absolute value of the sum of the regression coefficients (an ℓ_1 penalty), which will also shrink the coefficients as in ridge regression. The Lasso penalty can be written as $P^{\text{Lasso}}(\boldsymbol{\beta}) = \lambda \sum_{j=1}^p |\beta_j|$ and the Lasso loss function thus can be written as a penalized OLS loss function in matrix notation such that

$$L^{\text{Lasso}}(\boldsymbol{\beta}) = \|\mathbf{Y} - \mathbf{X}\boldsymbol{\beta}\|^2 + \lambda \mathbf{W}^T \boldsymbol{\beta} \quad (5)$$

where L is the loss function, $\mathbf{X}$ is an $n \times p$ design matrix for predictors, $\mathbf{Y}$ is an $n \times 1$ vector of responses, $\boldsymbol{\beta}$ is a $p \times 1$ vector of regression coefficients, $\mathbf{W}$ is a $p \times 1$ vector of ± 1 values that are congruent with the sign of the associated element of $\boldsymbol{\beta}$, and $\lambda \geq 0$ is the regularization parameter that controls the degree of shrinkage.

To clearly elucidate $P^{\text{Lasso}}(\boldsymbol{\beta})$, Equation (5) can be alternatively formulated as

$$L^{\text{Lasso}}(\beta_1, \dots, \beta_p) = \left\| \mathbf{Y} - \sum_{j=1}^p \mathbf{X}_j \beta_j \right\|^2 + \lambda \sum_{j=1}^p |\beta_j| \quad (6)$$

From Equations (5) and (6), it can be seen that when $\lambda = 0$, Lasso is identical to OLS. Unlike ridge regression, using a penalty based on the sum of the absolute values of the regression coefficients allows some regression coefficients to be shrunk to exactly zero. In effect, this means that Lasso can be used to select predictor variables as well as to obtain regularized regression coefficients. That is, variables that are estimated to be exactly zero are not retained in the model and are “selected out,” similar to how stepwise regression selects which variables to retain and which to discard. When multiple coefficients are shrunk to 0, the solution is deemed to be sparse. Furthermore, $P^{\text{Lasso}}(\boldsymbol{\beta}) = \lambda \sum_{j=1}^p |\beta_j|$ applies an equal amount of shrinkage across all regression coefficients, meaning that coefficients with larger magnitudes (and thus are potentially more important, depending on scale) are shrunk less (Hesterberg et al., 2008). This is in opposition to ridge regression, which shrinks coefficients proportional to their size and will result in coefficients with larger magnitudes being more severely shrunk compared to those with smaller magnitudes.

Although Lasso’s use of $P^{\text{Lasso}}(\boldsymbol{\beta}) = \lambda \sum_{j=1}^p |\beta_j|$ has the advantage of shrinking some estimates to exactly zero and performing variable selection, the presence of the absolute value function means that L^{Lasso} is not differentiable

and hence does not have a closed-form solution. This can be seen by the presence of $\mathbf{W}$ in Equation (5) but not in Equation (1) or Equation (4), both of which do have an analytic solution. $\mathbf{W}$ introduces 2^p possible combinations of sign patterns because the absolute value function in $P^{\text{Lasso}}(\boldsymbol{\beta})$ makes the sign of the elements of $\hat{\boldsymbol{\beta}}^{\text{Lasso}}$ ambiguous. That is, the magnitude of the coefficient can be estimated without much computational overhead but the absolute value function only gives magnitude and not direction, so ensuring that all coefficients have the proper sign requires more computational burden to sift through the possible permutations. Therefore, in order to minimize L^{Lasso} , optimization algorithms must be applied. Tibshirani (1996) first outlined a quadratic programming algorithm that required $n2^p$ computations, which is computationally demanding with even moderate values of p . Efron et al. (2004) provided the more general, computationally simpler least angle regression (LARS) framework from which Lasso can be obtained. LARS requires np^2 computations, which is much more tractable for values of p into the thousands.

Very recently, a significance test for Lasso coefficients that does not involve resampling procedures or splitting of the data (which can adversely affect power) has also been derived by Lockhart et al. (2014) and incorporates the adaptive nature by which predictors are retained or removed from the model as λ changes (this process may perhaps be unclear at this point but will be discussed in further detail in the next section). While the statistical theory behind this method is rather involved and is beyond the scope of this paper, conceptually Lockhart et al. address the problem inherent with model selection in that predictors move in and out of the model and are not fixed as traditional test statistic distributions assume. Their method derives the appropriate asymptotic null F distribution for a test statistic at the value of λ for which the regression coefficient is no longer equal to zero, allowing for inferential tests and p values for all variables being considered in the model. An experimental R package *covTest* was also released in July 2014 that implements this method and will be discussed in more detail at the end of this section.

Due to the abundance of studies in behavioral sciences that continue to employ stepwise procedures, it is worth mentioning that Efron et al. (2004) also note that LARS can be used strictly as a selection procedure whereby coefficient estimates are obtained by another estimation method after important predictors are selected. That is, the LARS algorithm selects the predictors that are non zero, but instead of using $P^{\text{Lasso}}(\boldsymbol{\beta})$ to shrink coefficients, estimation such as least squares or maximum likelihood is utilized only on the selected coefficients instead. This procedure has been referred to a post-Lasso estimation (the selection can be done with either Lasso or LARS but the coefficient estimates are not regularized) and is further advocated by Belloni and Chernozhukov (2013) and can be directly implemented in SAS Proc *GlmSelect* with the *Lsc coeffs* option in

the Model statement. This method can help bypass concerns about inappropriate test statistic distributions and reliance of p values or R^2 for selecting variables. That is, the difficulty in selecting appropriate predictors that results from untrustworthy p values with stepwise regression can be circumvented because LARS selects variables irrespective of p values. All predictors that remain after LARS can be considered to be important to retain, otherwise they would have been “selected out.” However, if the interest in p values lies beyond selecting relevant predictors, this method may not fully address that analytic context because post-Lasso p values calculated with standard methods will remain untrustworthy.

Methods to Select the Regularization Parameter λ

The selection of λ is vital in the process of modeling with Lasso as it directly controls the amount of shrinkage. Models can be fit for any value of λ but choosing the optimal value that provides generalizable estimates that address overfitting while not overly penalizing coefficients is an obvious priority. Two methods are commonly used to arrive at an appropriate value for λ : cross-validation and information criteria. Cross-validation is the method that is often used with Lasso because degrees of freedom must be calculated in order to use information criteria such as AIC or BIC, whereas degrees of freedom are not straightforward to calculate with Lasso because of the adaptive nature inherent with predictor selection. Using the number of predictors in the model has been suggested as an approximation for degrees of freedom with relative success (Zou, Hastie, & Tibshirani, 2007).

Ideally, the model would be fit to one set of data and tested on a second set of data to assess how well it performs. However ideal, this is not a realistic approach, especially in behavioral sciences where data are expensive and time consuming to collect. This process can be approximated, however, with cross-validation. Cross-validation splits the data into K different folds, typically 5, 10, or n . An extreme value for λ is selected (typically towards the upper extreme) and then data are then fit on $K - 1$ folds with Lasso and the estimated coefficients are then used to predict values in the k th fold. The process is then repeated for $k = 1, \dots, K$. For instance, if $k = 5$, the model is fit to the first four folds of the data combined, then the resulting coefficients are used to predict data from the fifth fold. The process is then repeated by estimating coefficients using fold one, two, three, and five and fitting the predicted model to fold four, and so on until all folds have served in each capacity.

A measure of fit (e.g., mean squared error for linear models) is recorded for each value of k which are then averaged and standard errors calculated for all folds for a particular value of λ . The value for λ is then changed and the process continues to iterate, typically for 100 values of λ . The mean

squared error (or another appropriate fit measure) is then inspected to determine for which value of λ the mean squared error is the smallest.⁵ Using the minimum value can at times result in small values for λ meaning that little shrinkage is conducted and over fitting may not be fully addressed, partially defeating the purpose of implementing Lasso in the first place. As an alternative, some studies suggest to select λ to be the largest value of λ that is within one standard error of minimum mean squared error value at the minimum λ (Friedman, Hastie, & Tibshirani, 2010; Waldmann, Mészáros, Gredler, Fuerst, & Sölkner, 2013).

A similar process can be used with information criteria. Models are fit to the complete data (there are no folds with information criteria) using many values of λ and information criteria (e.g., AIC, BIC) are calculated at each value of λ for $AIC = n \log(RSS) + 2df$ and $BIC = n \log(RSS) + df \log(n)$ where RSS is the residual sums of squares and df is the degrees of freedom. The λ value that produces either a local minimum (the default in SAS Proc GlmSelect) or the global minimum is then utilized as λ .

Overview of Lockhart et al. (2014) Method for Calculating Lasso p -values

As mentioned previously, p values from methods that select predictor variables adaptively (like stepwise regression) are often untrustworthy because the degrees of freedom and/or the standard errors are often incorrect and no method for obtaining values have been developed without resampling or splitting the data set (Lockhart et al., 2014; Thompson, 2001). The problem is no different in the context of Lasso although Wasserman and Roeder (2009) and Meinshausen and Bühlmann (2010) have developed methods for computing Lasso p values by splitting the dataset. However, the method introduced by Lockhart et al. does not require any resampling or splitting and thus uses the original data in its entirety.

Although the theoretical details are extensive, the general conceptual idea of the Lockhart et al. (2014) method is similar to a likelihood ratio test. In a standard likelihood ratio test, nested models can be compared by calculating the difference in the deviance ($\text{deviance} = -2 \log(\text{likelihood})$) between the full and restricted model, which can be shown to be χ^2 distributed with degrees of freedom equal to the difference in the number of parameters in the full and restricted models. Relatedly, Lockhart et al. show that the covariance between the observed values for the outcome ($\mathbf{Y}$) and the model predicted values for the outcome ($\mathbf{X}\hat{\beta}$) can serve a similar purpose.

More specifically, to inferentially test whether a predictor that enters the model at λ_j is equal to 0 in the population (for

$j = 1, \dots, p$ an index for λ values where a predictor enters the model), first let A be defined as the set of predictors in the model just prior to λ_j (Figure 2 and 3 in the next section may help visualize this process if it is unclear). The covariance of the predictors in set A (which does not include the predictor of interest) is calculated at λ_{j+1} , $Cov(\mathbf{Y}, \mathbf{X}_A \hat{\beta}_A)$, and is then subtracted from $Cov(\mathbf{Y}, \mathbf{X} \hat{\beta})$ for the model at λ_{j+1} (which includes the predictor of interest). λ_{j+1} is used instead of λ_j because the coefficient for the predictor that enters the model at λ_j will be essentially 0 at exactly λ_j (i.e., $\mathbf{X}_A \hat{\beta}_A = \mathbf{X} \hat{\beta}$ at λ_j). The test statistic T is then formed by dividing this difference by the estimate for the residual variance, σ^2 such that

$$T_j = \left(\langle \mathbf{Y}, \mathbf{X} \hat{\beta}(\lambda_{j+1}) \rangle - \langle \mathbf{Y}, \mathbf{X}_A \hat{\beta}_A(\lambda_{j+1}) \rangle \right) / \sigma^2 \quad (7)$$

This assesses how much of the covariance between the outcome variable and the predicted values is attributable to the predictor that entered the model at λ_j . Lockhart et al. (2014) then determine which F distribution is appropriate to ensure that the adaptive nature of the predictor selection is taken into account.

Similar to the likelihood ratio test, the Lockhart et al. (2014) method computes p values for each predictor without splitting the data and also without requiring standard errors to be calculated for each individual predictor as would be needed for a Wald-type test. To emphasize, the Lockhart et al. method does not calculate standard errors for the individual regression coefficients; it instead uses the change in the covariance to inferentially test whether the change in the covariance between the observed outcome variable and the predicted outcome variable is significantly different from 0 after including the predictor of interest.

APPLIED DATA EXAMPLE

To demonstrate both the shortcomings of stepwise selection and the problem of overfitting with standard OLS in general, we use data on test scores in California schools that appeared in Stock and Watson (2003) and also are included in the *Ecdat* package in R. Although regularization methods are most often applied to high-dimensional models (i.e., models with a very large number of predictors) in big data or biomedical applications, even with a fairly large sample and a fairly low-dimensional model, overfitting and over-selection of predictors can still present analytical problems (Subramanian & Simon, 2013). This example will highlight how the fairly low-dimensional problems common in behavioral sciences can benefit from regularization methods even though these disciplines do not always embody the traditional high-dimensional problems that are associated with regularization and Lasso in particular.

⁵Note that this is the average mean squared error across cross-validation samples for predicting the k th fold at a particular value of λ , not a single fit to the complete data. Otherwise, the OLS $\lambda = 0$ solution would have the lowest mean squared error.

Description of Data and Models

Data

The dataset contains mean math and reading test scores from 420 public school districts in California for the 1998–1999 school year. Test scores are an average from students between kindergarten and eighth grade, and 10 school characteristics are also included in the dataset. This results in a fairly large n to p ratio of 42.

The 10 predictors are (1) total enrollment in the district, (2) number of teachers in the district, (3) percentage of students qualifying for CalWorks,⁶ (4) percentage of students qualifying for free or reduced lunch, (5) the number of computers in the district, (6) the number of computers per student, (7) the expenditure per student (in thousands of dollars), (8) the ratio of teachers to students, (9) the per capita income for the district (in thousands of dollars), and (10) the percentage of English language learners in the district. The outcome measure used in the example is an arithmetic mean of reading and math scores which had a mean of 654.16 and a standard deviation of 19.05 for districts in the data.

Models

Model selection and the resulting regression coefficients will be compared between stepwise OLS and Lasso. For stepwise OLS, the criteria for retention in the model will be a significance level of $p < 0.05$.⁷ For Lasso, 10-fold cross validation will be used to select λ using the one standard error above the minimum criterion. P values for Lasso estimates using the method by Lockhart et al. (2014) will be compared to p values obtained from OLS when variables enter the model one at a time, in the same order. The Lasso analyses will be conducted in R using the `covTest` and `glmnet` packages and annotated R code is provided in the appendix. The appendix also contains SAS code, but R is solely discussed in-text because its open-source nature allows for the most up-to-date methods from the rapidly expanding literature on Lasso.

Results

Selecting a Value for λ

Using the `cv.glmnet` function in R, the value of λ that minimizes the mean squared error is 0.144 and the largest λ within one standard error (1SE) is 1.773 (these values are output directly by `glmnet`). The values for λ that are output by `glmnet` change slightly every time the program is run because they are calculated by cross-validation, which

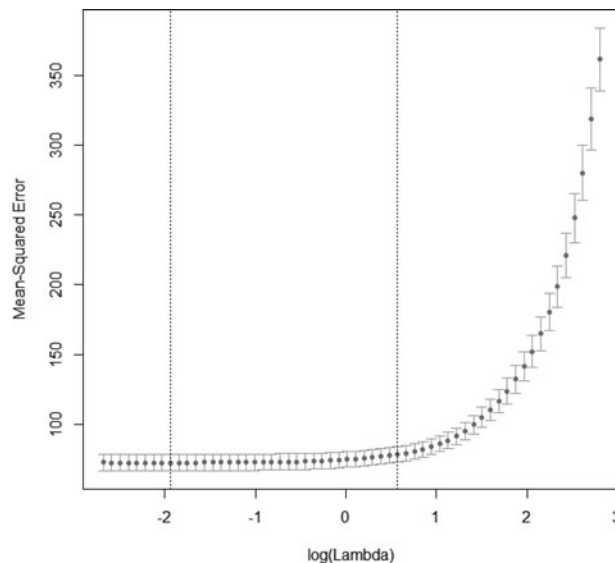


FIGURE 1 Mean squared error with standard error bars for k -fold cross validation as a function of $\log \lambda$. Vertical lines represent the value for $\log \lambda$ selected according to either the lowest mean squared error or 1 standard error above the lowest mean squared error criteria.

randomly splits the data into folds. Each time the procedure is run, the data are split differently, which will give slightly different results (this can be avoided by using a `set.seed` option in R; for the applied analysis the seed was set to 1028). Readers running the example code may not necessarily reach these exact values if a different seed value is used or if no seed is specified. Figure 1 shows a plot of the mean squared error (on the vertical axis) as output by R for various values

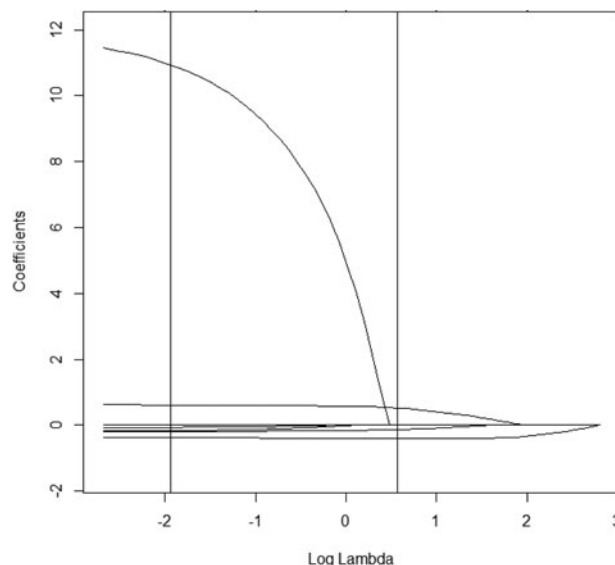


FIGURE 2 Unstandardized coefficient estimates for all candidate predictors as a function of $\log \lambda$ for full range of y-axis. Vertical lines represent the value for $\log \lambda$ selected according to either the lowest mean squared error or 1 standard error above the lowest mean squared error criteria.

⁶CalWorks is a program in California that provides resources to very low-income families. To qualify, California residents must be responsible for a child under 19 years old, have a very low income, and either work for very low wages, be unemployed, or be on the verge of unemployment.

⁷Recall that stepwise p values do not address the adaptive nature of the model and are often quite untrustworthy.

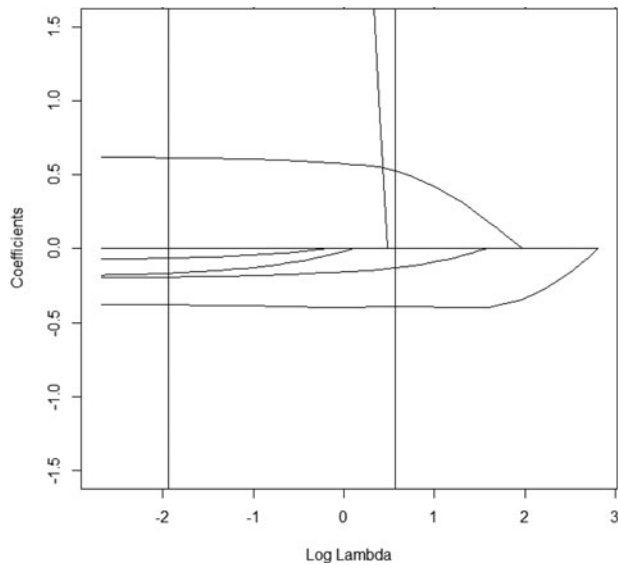


FIGURE 3 Unstandardized coefficient estimates for all candidate predictors as a function of $\log \lambda$ for restricted values of the y-axis. Vertical lines represent the value for $\log \lambda$ selected according to either the lowest mean squared error or 1 standard error above the lowest mean squared error criteria.

of λ (on the horizontal-axis; by default, `glmnet` uses $\log \lambda$). The imposed vertical lines in Figure 1 indicate the relative position of the minimum and 1SE values of $\log \lambda$. It can be seen that for large values of λ (which more severely shrink coefficients), the mean squared error is quite high because all regression coefficients are shrunk to zero, resulting in poor predictive power of the model. As values of λ decrease, the mean squared error begins to decrease because regression coefficients are not shrunk all the way to zero, which results in better predictive power. Although the plot appears to flatten as λ becomes increasingly small, the mean squared error actually begins to increase for values of λ smaller than 0.144 ($\log[0.144] \approx -1.94$). This effect is typically much more noticeable than the slight increase in the example provided here.

Prior to finalizing a value for λ , inspecting a plot of coefficient values as a function of λ is also helpful. Figure 2 and Figure 3 plot $\log \lambda$ on the horizontal axis and penalized unstandardized regression coefficients on the y-axis (one coefficient had a much larger magnitude than the others so Figure 2 shows the full range of the y-axis and Figure 3 truncates the range of the y-axis for a more nuanced inspection of the other 11 predictors). In Figures 2 and 3, each line represents an individual candidate predictor in the model and the lines show how the coefficient estimates change across values for λ . For large values of λ , it can be seen that all coefficients are equal to zero because the penalization (and thus the degree of shrinkage) is extremely large. As λ decreases, important predictors begin to be nonzero and, for small values of λ , all coefficients are non zero as they would almost certainly be if the model was fit with OLS. The two imposed

TABLE 1
Coefficient Estimates and Model Fit for Stepwise Selection and Lasso

Predictor	Stepwise OLS	Lasso
Intercept	654.97	665.67
Percentage free or reduced lunch	-0.40	-0.39
Average per capita income	0.62	0.53
Percentage english language learner	-0.19	-0.13
Annual expenditure per student	1.69	—
Number of computers per student	13.51	—
R^2	0.81	0.79
Mean squared error	8.41	8.72

OLS = Ordinary Least Squares

vertical lines (not output by default but included in the code in the appendix) serve as a reference for the minimum and 1SE values of $\log \lambda$.

Regression Coefficients

Regression coefficient estimates for stepwise OLS and Lasso are provided in Table 1 along with the R^2 and mean squared error for the model. Differences between the models are fairly noticeable even though there are 42 observations per predictor in the model. If the number of observations per predictor were decreased, the differences between methods would likely grow much larger. The sparse Lasso solution yielded three non zero predictors (percentage free/reduced lunch, average per capita income, and percentage English language learner) using λ 1SE from the minimum. On the contrary, stepwise OLS selected five predictors (percentage free/reduced lunch, average per capita income, percentage English language learner, number of computers per student, and district expenditures per student). The coefficient estimates are also somewhat different as well. The estimate for percentage free/reduced lunch is fairly close between methods because this was the predictor with the largest magnitude, meaning that Lasso shrinkage is minimal in this case (Hesterberg et al., 2008). As coefficients possess less magnitude, the shrinkage applied with Lasso becomes relatively greater: the coefficient estimate for average per capita is 12% smaller with Lasso and the coefficient estimate for percentage English language learner is 25% smaller with Lasso. Although the model R^2 values are not vastly different as would be anticipated by a simulation by Babyak (2004) because of the large n to p ratio, the R^2 for Lasso is slightly smaller (79% vs. 81%). Also as anticipated, the mean squared error for Lasso is higher (8.72 vs. 8.41) than for stepwise selection because Lasso is purposefully introducing a controlled bias into the estimate. However, this small amount of bias makes the estimates more generalizable and the model more parsimonious.

Table 2 shows the p values as calculated by OLS with stepwise selection constrained to include all predictors and

TABLE 2
Comparison of OLS and Lasso p Values for Predictors
Entering the Model One at a Time

Predictor	P value	
	Lasso	Stepwise
Total enrollment	0.70	0.54
Number of teachers	0.95	0.82
Percentage eligible for CalWorks	0.19	0.17
Percentage free or reduced lunch	< 0.01	< 0.01
Number of computers	0.98	0.81
Number of computers per student	0.89	0.01
Annual expenditure per student	0.21	0.02
Student-teacher ratio	0.89	0.50
Average per capita income	< 0.01	< 0.01
Percentage english learner	< 0.01	< 0.01

Note. Lasso p values are calculated with the experimental covTest package in R using the Lockhart et al. (2014) method.

The distribution for the Lasso p values is $F_{(2,410)}$.

as calculated by the covTest package in R using the method proposed by Lockhart et al. (2014). Stepwise selection was constrained to select all predictors so that predictors would enter the model in order of importance similar to Lasso and because the method in Lockhart et al. outputs p values for all candidate predictors. Thus, a side-by-side comparison between OLS p values and Lasso p values obtained through the Lockhart et al. method can be conducted. The p values as calculated by OLS are noticeably smaller than those obtained with Lasso, particularly for the number of computers per student (Lasso $p = 0.89$ vs. stepwise $p = 0.01$) and annual expenditure per student (Lasso $p = 0.21$ vs. stepwise $p = 0.02$) predictors.

Lasso Assumptions and Assumption Tests

Lasso is a method by which predictor variables are more appropriately selected (compared to competing model selection procedures, e.g., stepwise regression) and whose coefficient estimates are more generalizable to the population; however, the interpretation and formulation of the regression model remains mostly unaffected. Regression coefficients are regularized to account for potential overfitting, but the interpretation of regression coefficients is not appreciably altered. Further, model assumptions are identical between models whose coefficients are estimated with OLS (or maximum likelihood, etc.) and Lasso. For metric outcomes, this includes normality of the residuals, homoscedasticity of the residuals, independence of the residuals, and the proper functional form and inclusion of all relevant predictors,⁸ all of which can be in-

spected as they would were the coefficients estimated without Lasso.

The residuals for the Lasso model in the applied data example were approximately normally distributed in the applied example as adjudicated by a histogram, a Kolmogorov-Smirnov test, and Shapiro-Wilk test. The homoscedasticity assumption appeared reasonable when inspected by plotting the Lasso residuals against the Lasso predicted values. Plots of the Lasso residuals with each individual predictor are shown in Figure 4 with a loess line superimposed in gray. Figure 4 indicates that some non linearity may be present with the average per capita income predictor. A quadratic term for average per capita income was added to the model but it was not selected with either Lasso or with stepwise regression so the linearity assumption appears to be fairly reasonable.

DISCUSSION

Although Lasso is a booming research area in theoretical and computational statistics, behavioral science researchers have still yet to implement Lasso and exploit its advantages (Johnson & Sinharay, 2011), even as software for fitting such models has become increasingly accessible and available. The example discussed in this paper featured a very large ratio of observations to predictors and differences between OLS and Lasso were still quite noticeable. Regression applications in behavioral sciences routinely have small or moderate samples with n to p ratios that are likely smaller than the provided example although the more widespread use of fMRI can result in scenarios that closely mirror those for which Lasso was originally intended. Utilizing standard OLS techniques in such situations may result in estimates and interpretations that potentially suffer from overfitting or from over-selection of predictor variables through stepwise selection and/or reliance on p values, which ultimately may hurt the authenticity, parsimony, and generalizability of conclusions. Straightforward software exists for implementing more modern methods such as Lasso that do not overselect predictors, mitigate overfitting through regularization, are beginning to incorporate inferential tests and p values that are commonly reported in behavioral sciences, and provide familiar fit measures such as R^2 , AIC, and BIC.

Beyond straightforward OLS-type problems upon which this expository introduction focused, Lasso can also be applied to a wealth of other modeling contexts that are routinely encountered in behavioral sciences. Lasso can be applied to a wide range of dependent variable measurement scales that encompass the generalized linear model including binary, Poisson, multinomial, and time-to-event models (Gui & Li, 2005; Tibshirani, 1996, 1997). All of these distributions can be fit in the glmnet package in R with additional options

⁸Of course, real-world data never perfectly conform to the model assumptions and, as a result, the residuals from the estimated model often exhibit a small degree of correlation (e.g., trivial nestedness of the data, minor model misspecification due to the unavailability of a predictor in the data). Minor departures from these assumptions do not impact the estimation or

interpretability of the model (Cohen et al., 2003; Lomax & Hahs-Vaughn, 2013).

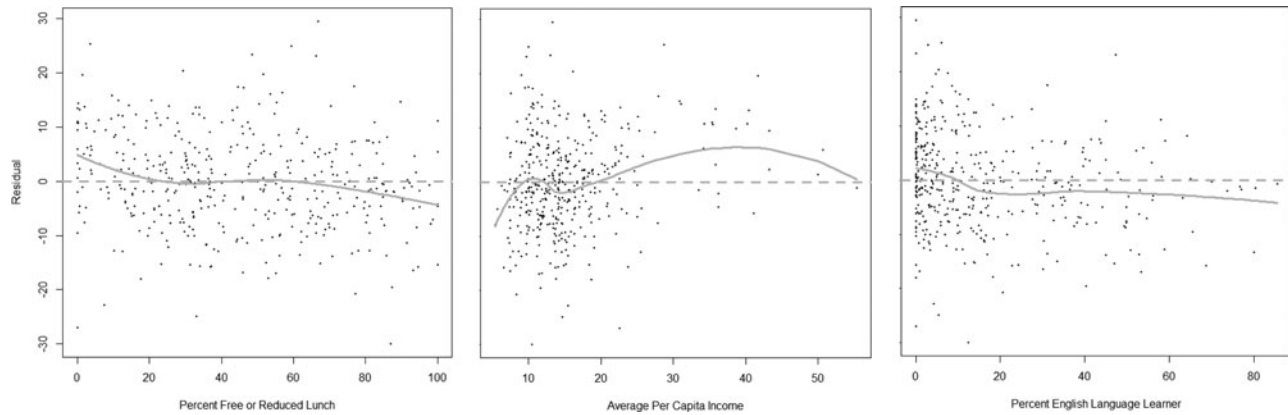


FIGURE 4 Lasso residuals plotted against predictor variable values with superimposed loess line. Dotted line is 0-line.

that require only a few additional words of code. Of particular interest to behavioral science researchers, Lasso can also be implemented for fixed effects in linear and generalized linear mixed models (aka HLM or HGLM, respectively; Groll & Tutz, 2014; Park & Hastie, 2007; Schelldorfer, Meier, & Bühlmann, 2014) and also for mixture models with metric outcomes (Khalili & Chen, 2007; Städler, Bühlmann, & Van de Geer, 2010). The `glmLasso` package in R can implement Lasso for generalized linear mixed models (Schelldorfer et al., 2014).

As one drawback with Lasso that is of interest to behavioral science researchers, standard Lasso operates on all predictors variables simultaneously and cannot explicitly accommodate multiparameter factors. For instance, if ethnicity is a desired factor in the model and it is dummy coded into categories (e.g., White, Black, Hispanic, Asian, and Other), standard lasso will treat the four dummy coded variables as distinct predictors rather than selecting and penalizing them as a single, multi parameter factor. If a particular reference group is not explicitly of interest, this may not be problematic. For instance, perhaps the dummy variable for White is the only predictor selected in the model from the above parenthetical example; the reference group will change to include Black, Hispanic, Asian, and Other individuals rather than the group that was pre specified to be the reference group. Yuan and Lin (2006) first addressed the problem of selecting sets of predictors (e.g., multiparameter factors, linear and quadratic components) with metric outcomes with the group Lasso that allows set of parameters to be selected together and also penalizes them together. More specifically, the factor as a whole is penalized using Lasso and then a ridge-type penalty is applied to the regression coefficient within a group. Meier et al. (2008) later extended group Lasso to logistic regression. The `grplasso` R package can implement group Lasso for metric, binary, and Poisson outcomes (Meier, 2009). The `glmmlasso` package also allows users to apply group Lasso.

Although Lasso was the near exclusive focus of this paper, readers should note that there are other regularization

methods that serve similar purposes. Zou (2006) discussed how standard Lasso does not asymptotically select the appropriate predictors under all conditions and then proposed Adaptive Lasso that adaptively weights the penalty term. This method was shown to perform best among competing methods in a simulation in Zou and is available in both SAS `Proc GlmSelect` and the `parcor` R package. The Elastic Net applies penalties from both ridge regression and Lasso, which can help when predictors are highly correlated. With highly correlated predictors, Lasso tends to select one predictor from the set of correlated predictors whereas the Elastic Net tends to select the correlated variables together (Zou & Hastie, 2005). When the number of predictors p is much larger than the number of observations n , the Dantzig Selector is another method of variable selection that applies the same penalty as Lasso except it is applied to the residuals rather than a likelihood or least squares function (Candès & Tao, 2007). Research scenarios where p is much greater than n are not commonplace in the behavioral sciences outside of fMRI or genetics research, however.

CONCLUDING REMARKS

The popularity of Lasso as a method to efficiently model high-dimensional data in other disciplines such as statistical genetics and big data has exploded in recent years. Although the methodological problems in these fields do not always directly correspond to problems in the behavioral sciences that are often of much lower dimension, as shown in the low-dimensional example, Lasso can provide more parsimonious predictor selection and more generalizable inferences than commonly implemented methods like stepwise selection or standard OLS estimation. As software for Lasso and regularization continues to increase both in volume and accessibility, we hope that behavioral science researchers will incorporate Lasso into their methodological toolboxes as researchers in other disciplines have done recently.

ARTICLE INFORMATION

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Ethical Principles: The author affirms having followed professional ethical guidelines in preparing this work. These guidelines include obtaining informed consent from human participants, maintaining ethical treatment and respect for the rights of human or animal participants, and ensuring the privacy of participants and their data, such as ensuring that individual participants cannot be identified in reported results or from publicly available original or archival data.

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SUPPLEMENTAL MATERIAL

Supplemental data for this article can be accessed on the publisher's website.

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